MIL-M-13306A(Ord)
9 June 1961
SUPERSEDING
MIL-M-13306 (Ord)
17 March 1954

MILITARY SPECIFICATION

MOUNT, TELESCOPE: M96

1. SCOPE

1.1 This specification covers one basic type of telescope mount primarily designed to provide for compensation of angles of cant for direct fire with the 155 MM Self-Propelled Howitzer M44.

2. APPLICABLE DOCUMENTS

2.1 The following documents or the issue In effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-F-14619	Fire Control Parts, Equipment and Tools; Packaging of
MIL-I-45208	Inspection Requirements, General *Specification For*

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property

FSC 1240

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DRAWINGS

Ordinance Corps

F7654413 Mount Telescope: M96

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production one mount, representative of production, shall be made available to the Government to determine if the mount meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.3 Construction.- The mount shall be manufactured in conkace with Drawing F7654413, "Mount, Telescope: M96" and drawings pertaining thereto. The requirements of this specification are detailed only to obtain the desired mechanical characteritics, performance % and pemanence to insure inherent quality and performance capabilities prior to final cceptance by the Government.

3.4 General specifications.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence
(b) Protective finished surfaces
(c) Spectal accuracy.
(d) Finish painting and protective
(e) Electrical
(f) Workmamship

3.5 Environmental.

3.5.1 Operating temperature.- The mount shall meet the applicable requirements of 3.6.5 while exposed and thermally stabilized at ambient temperatures of plus 150 degrees and minus 40 degrees Fahrenheit subsequent to which the mount shall meet the requirements of 3.5.2.

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3.5.2 Vibration.- The mount positioned and supported in its using position shall be vibrated at a constant frequency of thirty cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of minutes plus or minus fifteen seconds. Subsequent to vibration there shall be no evidence of damage or physical failure and the mount shall meet the requirements of 3.6

3.5.3 Sealing.- The mount shall meet the requirements of Procedure I of the rain test of Specification MIL-F-13926.

3.6 Adjustments.- With the mount positioned with the flat mounting surfaces in a horizontal plane the mount shall meet the requirements of 3.6.1 to 3.6.4 inclusive.

3.6.1 Level vial reflection mirror.- The line of sight to the level vial reflection mirror of the cant correction mechanism shall be as indicated on Drawing F7654413.

3.6.2 Telescope clamp assemblies.- The alignment of the telescope clamp assemblies shall be as indicated on the drawings within 0.001 inch.

3.6.3 Cant correction travel.- The cant correction mechanism shall cover an excursion of at least 15 degrees clockwise and 15 degrees counterclockwise from midpoint position of the level vial.

3.6.4 Backlash.- Backlash between the level vial and the reticle rotation shaft as measured at the coupling at standard ambient temperature, shall not exceed thirty minutes when a torque of five pound inches is applied to the coupling in each direction.

3.6.5 Operability .- the cant correction knob shall not exceed 7 pounds checked at standard ambient temperatures and shall not exceed 9 pound inches when checked at extreme temperatures of 3.5.1.

3.7 Illumination.- With the instrument light energized with 3 volts, the level vial markings shall be plainly when observed in a darkened area.

3.8 Identification marking.- Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General provisions.- The quality assurance Provisions of referenced documents to the extent specified herein, together with provisions of this section, are applicable to inspection performed by the contractor and for Government verification.

4.1.1 The contractor shall provide an inspection system in accordance with Specification MIL-I-45208.

4.2 Production model.- Unless otherwise specified in the contract, one mount shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract, The mount shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein,

4.2.1 Production model failure.- Should the production model fail to pass the prescribed tests, the contractor shall those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.3 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.3.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.4 Inspection of components.- All positional and function characteristics, shown on the drawings contained in the contract, which indicate precision requirements and have a bearing on functional accuracy and performance of the mount shall inspected in accordance with the written inspection system required by Specification MIL-I-45208. These characteristics are defined as drawing requirements affecting concentricity, squareness, parallelism, microfinishes 64 RMS or better, threads, identification marking, and performance requirements specified in notes on the drawings and in separate performance specifcations referenced on the drawings.

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Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with Table I.

TABLE I - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	25	0	1
36 - 50	30	0	1
51 - 150	30	1	2
151 - 300	60	2	3
301 - Over			

4.4.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.5 Material.- Control for requirements of material shall be in accordance with Specification MIL-I-45208.

4.6 Inspection equipment.- Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

4.6.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specification tolorance, except as otherwise specified. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be Inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.7 Tests.

4.7.1 Classification of defects.- The classification of defects in Tables II and III shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

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4.7.1.1 Disposition of rejected lots. Rejected lots *shall be screened* for all defective characterist. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 1.5%	Req't *Par.*	QA. PAR.	Test Method
10l. Adjustments	3.6	4.7.3	StandardMeas-ment (MU)
102, Telescopoe assemblies clamp	3.6.2	4.7.5	SME
103. cant correction	3.6.3	4.7.6	SME
104. Backlash	3.6.4	4.7.7	SME
1050 0perability (standard ambient temperature)	3.6.5	4.7.8	SME-DST-(Device, Special Testing)

MINOR: AQL 4.0%	Req't Par.	QA. PAR	Test Method
201. Level vial reflection mirror	3.6.1	4.7.4	Visual
202. Illumination	3*7	4.7.9	Visual
203. Identification marking	3.$	4.8	Visual

TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level L7 *and Sampling*

CRITICAL. NONE DEFINED

MAJOR: AQL 4.0%	Req't Par.	QA. Par.	Test Method
101. Operating temperature (operability)	3.5.1 3.6.5	4.7.2.1	Device, Spe-(DST)
102. Vibration	3.5.2	4.7.2.2.	DST
103, Sealing	3.5.3	4.7.2.3	DST

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TABLE III - CLASSIFICATION OF DEFECTS Cont'd

MINOR: NONE DEFINED

4.7.2 Environmental.

Operating temperature.- The mount being tested shall be installed in the test chamber in a manner similar to its normal operating position. The mount shall then be tested in accordance with Procedure I, condition C-G of Specification MIL-F-13926. While exposed and thermally stabilized at the above conditions, the mount shall be tested as outlined in 4.7.8. The mount shall then be thermally stabilized at room temperature and tested as specified in 3.5.1.

4.7.2.2 Vibration.- The mount shall be positioned on a vibration fixture in its normal operating position. The vibration fixture shall be capable of producing a constant frequency with an amplitude and for the duration as specified. Upon completion of the vibration, the mount shall be visually examined for evidence of damage and shall meet the requirements specified in 3.5.2.

4.7.2.3 Sealing.- Upon completion of the sealing test specified in 3.5.3 the mount shall be visuallyexamined for evidence of water penetration or damage to the cant correction assembly of the mount.

4.7.3 Adjustments. - The mount shall be positioned with the flat mounting surfaces placed on a leveled surface plate which will locate the mount in the horizontal plane as specified in 3.6.

4.7.4 *Level* vial reflection mirror.- With the mount positioned as specified in 4.7.3 and as shown on Drawing F7654413, the line of sight and eye distance shall be determined by standard measuring equipment *and* visual means for compliance with 3.6.1.

[illegible] surface plate the parallelism of the telescope clamp assemblies indicated on Drawing F7654413 shall be determined by means of an aligning bar and a vernier height gage with dial indicator attached. The bar shall be placed in the telescope clamp assemblies. The dial indicator shall be placed on one end of the bar. Deflection of the dial indicator, when moved along the length of the bar, shall be within the limits specified in 3.6.2.

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4.7.6 Cant correction travel.- *The mount* shall *be* positioned on a sine plate of 15 degrees. The knob shall be rotated until the two indicates coincide. The reticle rotation shaft slot shall be In a horizontal plane with respect to the indices as measured by means of a dial indicator moved along the length of the slot. The knob shall then be rotated in a clockwise direction until the level vial bubble is centered. The sine plate shall then be rotated 1$0 degrees and the mount placed upon it. The knob shall then be turned in a counterclockwise direction until the level vial bubble is centered. This will indicate that the 15 degrees *clockwise and* counterclockwise movement has been attained.

4.7.7 Backlash.- The mount shall be petitioned as specified in 4.7.3. The level vial bubble shall be centered and the coupling slot in a horizontal plane as determined by a dial indicator placed in contact with the bottom of the slot and the reading noted. A torque wrench of five pound inches shall then be attached to the coupling and rotated clockwise then counterclockwise to center. The reading on the dial indicator shall be noted. The target~@&~ench shall then be rotated counterclockwise then clockwise center. The reading on the dial indicator shall be noted. The difference *in* readings shall not exceed 0.0087 inch, which is *equivalent* to thirty minutes of arc in a circle of two Inches in diameter, which is the diameter of the coupling.

4.7.8 Operability *A* calibrated torque wrench and an adapter shall be attached to the cant correction knob . The measured value of torque, as read directly *on* the torque wrench, shall not exceed the limits specified when tested at room temperature and at the extreme temperatures specified in 3.5.1.

4.7.9 Illumination.- A 3 volt source shall be applied to the instrument light assembly in a darkened area. When observed, the level vial markings shall be plainly visible as specfied.

be visually inspected for compliance with Standard MIL-STD-130.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Preservation, packaging, packing and marking shall be in accordance with SpecificationMIL-F-14619/P12729. The level of protection shall be as specified in the procurement document .

6. NOTES

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6.1 Intended use.- The Telescope Mount: M96 is primarily intended to support the Elbow Telescope:'"Pm M93 in the l15MM Self-Propelled Howitzer: M44. The mount provides cant correction for the Elbow Telescope: M93.

6.2 Order data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of an applicable level of preservation, packaging and packing
(c Applicable stock number
(d) Provisions for production model including date of delivery (see 3.1)

6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon specific request of the contractor.

6.4 Definitions.

6.4.1 Special testing *de*
are defined as special inspection
standard measuring equipment.

6.4.2 Standard measuring equipment (SME).- Standard measuring equipment is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also Includes commercial testing equipment such as meters, optical comparators, and so forth.

NOTICE: When Government drawings, specifications, or other data are *used* for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligations whatsoever; and the fact that the Government may have formulated, furnished, or in anyway supplied the said drawings, specifications, or other data is not to be regarded by implication *or* otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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